

RUICHUN MA

Senior Researcher, Microsoft Research - Vancouver

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RESEARCH INTERESTS

Wireless networking and sensing, Metasurface-based wireless systems, Embodied & Agentic AI

EDUCATION

Yale University

2020 - 2024

Ph.D. in ECE

Advisor: Wenjun Hu, Lin Zhong

Thesis: Bridging Heterogeneous Wireless Networks with Metasurfaces

[SIGMOBILE Doctoral Dissertation Award](#) (One awardee worldwide per year)

University of Science and Technology of China

2016 - 2020

School of the Gifted Young (Rank:1/28)

Bachelor of Engineering in EE & AI

PUBLICATIONS

(* marks interns mentored by me.)

[Preprint] Heyu Guo*, Shanmu Wang*, [Ruichun Ma](#), Shiqi Jiang Yasaman Ghasempour, Omid Abari, Baining Guo, Lili Qiu. **OmniVLA: Physically-Grounded Multimodal VLA with Unified Multi-Sensor Perception for Robotic Manipulation**. Arxiv [\[pdf\]](#)

[ToN'25] [Ruichun Ma](#) and Wenjun Hu. **Cross-Medium Networking with Transflective Flexible Metasurfaces**. IEEE Transactions on Networking, 2025. [\[pdf\]](#)

[MobiCom'25] Hao Pan, Yezhou Wang, Jiting Liu, [Ruichun Ma](#), Lili Qiu, et al. **CGMM: Non-Invasive Continuous Glucose Monitoring in Wearables Using Metasurfaces**. In the International Conference on Mobile Computing and Networking, 2025.

[SenSys'25] [Ruichun Ma](#), Yasuo Morimoto, John Ho, Sam Shiu, Jiang Zhu. **mmET: mmWave Radar-Based Eye Tracking on Smart Glasses**. In the ACM Conference on Embedded Networked Sensor Systems, 2025. [\[pdf\]](#)

[HotNets'24] [Ruichun Ma](#), Lili Qiu, Wenjun Hu. **SurfOS: Towards an Operating System for Programmable Radio Environments**. In the ACM Workshop on Hot Topics in Networks, 2024. [\[pdf\]](#)

[MobiCom'24] [Ruichun Ma](#), Shicheng Zheng, Hao Pan, Lili Qiu, et al. **AutoMS: Automated Service for mmWave Coverage Optimization using Low-cost Metasurfaces**. In the International Conference on Mobile Computing and Networking, 2024. [\[pdf\]](#)

[MobiCom'24] [Ruichun Ma](#) and Wenjun Hu. **RF-Mediator: Tuning Medium Interfaces with Flexible Metasurfaces**. In the International Conference on Mobile Computing and Networking, 2024. [\[pdf\]](#)

[MobiCom'24] Yezhou Wang, Hao Pan, Lili Qiu, Linghui Zhong, Jiting Liu, [Ruichun Ma](#), et al. **GPMS: Enabling Indoor GNSS Positioning using Passive Metasurfaces**. In the International Conference on Mobile Computing and Networking, 2024.

[MobiCom'23] [Ruichun Ma](#), R. Ivan Zelaya, and Wenjun Hu. **Softly, Deftly, Scrolls Unfurl Their Splendor: Rolling Flexible Surfaces for Wideband Wireless**. In the International Conference on Mobile Computing and Networking, 2023. [\[pdf\]](#)

[HotNets'21] R. Ivan Zelaya, [Ruichun Ma](#), and Wenjun Hu. **Towards 6G and Beyond: Smarten Everything with Metamorphic Surfaces**. In the ACM Workshop on Hot Topics in Networks, 2021.[\[pdf\]](#)

RESEARCH EXPERIENCE

Microsoft Research Asia, Vancouver

Senior Researcher

Nov 2024 - Now

Manager: Dr. Lili Qiu

- Working on wireless systems for healthcare, embodied AI, and agentic system design.

Yale University

Research Assistant (PhD student)

2020 - 2024

Advisor: Dr. Wenjun Hu

- Designed metasurfaces as a cross-layer hardware tool to support heterogeneous wireless networks
- Built end-to-end systems via hardware-software co-design for next-gen wireless infrastructure
- Related research works are published in MobiCom, HotNets, ToN

Meta Platforms, Reality Labs

Research Scientist Intern

May - Nov 2024

Managers: Dr. Jiang Zhu

- Designed and built the first mmWave radar-based eye tracking glasses for AR/VR
- Proposed a tailored ML model and a novel pre-training strategy based on contrastive learning

Microsoft Research Asia, Shanghai

Research Intern

Jan - Aug 2023

Manager: Dr. Lili Qiu

- Designed novel metasurfaces for wireless coverage and sensing projects (two patents submitted)
- Developed an automated service framework for metasurface-aided mmWave coverage (MobiCom'24)
- Demonstrated metasurface-based wireless glucose sensing to Mr. Bill Gates

University of Illinois Urbana-Champaign

Research Intern

Jul - Oct 2019

Advisor: Prof. Haitham Hassanieh

- Developed a wireless mesh network protocol design, reducing the average packet delay by 30%

Imperial College London

Visiting student

August 2018

Advisor: Prof. Guang-Zhong Yang

- Developed an intelligent medical robot arm prototype for surgery (Runner-up award)

AWARDS & HONORS

[SIGMOBILE Doctoral Dissertation Award](#)

2025

Dissertation with Distinction Award at Yale

2025

Distinguished Graduate Scholarship (top 5%)

2020

Merit Student Scholarship (top 5%)

2017-2020

SERVICE

Technical program committee member:

MobiCom'26, MobiSys'26, SenSys'26, EWSN'25

Invited Reviewer:

IEEE TMC, IEEE TON, IEEE TOSN, ICASSP

TEACHING EXPERIENCE

Neural Networks and Learning Systems

Teaching Assistant, 2021 Fall @ Yale

Digital Signals and Systems

Teaching Assistant, 2019 Fall @ USTC